📝 DAY 1: STUDENT QUICK REFERENCE & OFFICIAL MCQ BANK

1-Page Cheat Sheet + 50 Official Pre-Assessment Questions (GLA University)

Course Code: GLA-HPC-BCSG701 | **Track:** Intel & NEC Supercomputing Alliance
Lead Trainer: Shivam Gupta | **Target Audience:** 60 B.Tech / M.Tech CS Students
Primary Reference: `Intel & NEC - High Performance Computing.pdf`

📌 PART 1: 1-Page Student Command Cheat Sheet

> **TIP**
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> ### ⚡ 2-MINUTE INSTANT-RECALL MASTERY BOX (80-90% IMMEDIATE KNOWLEDGE)
>
> - **HPC Core Physics:** Aggregates compute nodes across high-speed interconnects to bypass single-machine RAM bandwidth bottlenecks (Memory Wall).
> - **Hardware Triad:** Intel Xeon (Host Orchestration & FP64 math) + Intel GPU Max (Dense Matrix Parallelism) + Habana Gaudi (Tensor TPCs & On-Die 24x 100GbE RoCE).
> - **Linux Golden Rule:** Everything is a file starting at root `/`. Paths are Absolute (`/path`) or Relative (`./path`).
> - **Top 5 Life-Saving Commands:**
>   1. `lscpu | grep -E 'Model name|avx512|amx'` (Inspect CPU model & vector/matrix engines)
>   2. `mkdir -p ~/project/{src,bin,logs}` (Create complete clean directory tree in 1 command)
>   3. `chmod +x script.sh` (Add executable permission bit; fixes `Permission denied`)
>   4. `ssh-keygen -t ed25519 -N ""` (Generate modern high-security elliptic curve SSH key)
>   5. `source /opt/intel/oneapi/setvars.sh` (Initialize Intel oneAPI environment, `$PATH` , and compilers)

1. Navigation & Inspection

Command	What It Does	Example
`pwd`	Print working directory	`pwd` -> `/home/student/hpc`
`cd <path>`	Change directory	`cd ..` (up), `cd ~` (home), `cd -` (previous)
`ls -la`	List all files with permissions & hidden files	`ls -lh` (human readable sizes)
`lscpu`	Display CPU microarchitecture & flags	`lscpu
`free -h`	Display available RAM	`free -h`

2. File & Directory Operations

Command	What It Does	Example
`mkdir -p`	Create nested directory tree	`mkdir -p project/{src,data,logs}`
`touch`	Create empty file	`touch main.cpp`
`cp -r`	Copy file or recursive folder	`cp -r src/ backup_src/`
`mv`	Move or rename file	`mv old.txt new.txt`
`rm -r`	Remove file or folder (*CAUTION: Permanent!*)	`rm -rf temp_logs/`
`cat` / `head` / `tail`	View file contents	`head -n 20 data.csv`
`grep -i`	Search text in files	`grep -i "error" output.log`

3. Permissions & Shell Automation

Command / Syntax	What It Does	Example
`chmod +x`	Add executable permission to script	`chmod +x run.sh`
`#!/bin/bash`	Shebang: Tells OS to use Bash interpreter	First line of every `.sh` file
`VAR="val"`	Set variable (*NO SPACES around* `=`)	`THREADS=16`
`echo ${VAR}`	Read variable	`echo "Threads: $THREADS"`
`>` vs `>>`	Output redirection (overwrite vs append)	`cmd > out.log` vs `cmd >> out.log`
`set -e`	Exit script immediately if any command fails	Placed at top of robust scripts

4. SSH Remote Access & oneAPI

Command / Syntax	What It Does	Example
`ssh-keygen -t ed25519`	Generate secure SSH key pair	`ssh-keygen -t ed25519 -N ""`
`ssh user@host -p 22`	Connect securely to remote HPC node	`ssh shivam@10.100.1.5`
`source setvars.sh`	Initialize Intel oneAPI compilers & paths	`source /opt/intel/oneapi/setvars.sh`
`icpx --version`	Verify Intel oneAPI C++ Compiler	`icpx --version`

🎯 PART 2: Day 1 Official Study Checklist (22 Items)

- ▢ 01. HPC concept and purpose
- ▢ 02. HPC system overview
- ▢ 03. HPC versus ordinary computing
- ▢ 04. Intel Xeon Scalable Processors
- ▢ 05. Intel Data Center GPUs
- ▢ 06. Habana Gaudi AI Accelerators

- ▢ 07. Silicon hardware roles in HPC
- ▢ 08. Linux filesystem hierarchy & paths (absolute vs relative)
- ▢ 09. Core Linux navigation & file commands
- ▢ 10. Linux environment variables & `$PATH`
- ▢ 11. Shell script creation & execution permissions
- ▢ 12. Shell variables and command execution
- ▢ 13. Shell task automation workflows
- ▢ 14. SSH remote access purpose
- ▢ 15. SSH connection syntax & ports
- ▢ 16. Password vs Key-based authentication (`ed25519`)
- ▢ 17. Intel oneAPI Toolkit purpose
- ▢ 18. oneAPI components overview (DPC++/SYCL, oneMKL, oneDNN)
- ▢ 19. oneAPI environment access (`setvars.sh`)
- ▢ 20. Intel HPC Toolkit introduction
- ▢ 21. HPC compilation workflow (`icx` , `icpx` , `ifx`)
- ▢ 22. Intel HPC Toolkit tools (Intel MPI, Inspector, ITAC)

📝 PART 3: 50 Official Evaluated Pre-Assessment Questions (With Answers & Explanations)

> **Official 50-Question Diagnostic Bank for Day 1 Pre-Assessment (GLA University BCSG701)**
> **Track:** Intel & NEC Supercomputing Alliance × Edulateral Foundation
> **Gradation:** Easy (14) | Medium (21) | Hard (15) — Strictly structured across Day 1 Syllabus Modules 1 to 7.

🔷 Module 1: Introduction to HPC

Q01. [Easy] Which performance metric is the primary benchmark utilized to evaluate and rank the world's most powerful supercomputers on the official TOP500 list?

- A) Clock frequency measured in Gigahertz (GHz)
- B) Floating-point Operations Per Second (FLOPS) achieved on High Performance LINPACK (HPL)
- C) Input/Output Operations Per Second (IOPS) on NVMe storage
- D) Instructions Per Cycle (IPC) on integer ALU workloads **Answer: B** | *Explanation: The TOP500 supercomputing ranking measures the maximum sustained floating-point execution rate (Rmax in FLOPS), benchmarked using High Performance LINPACK (HPL) solving a dense system of linear equations.*

Q02. [Medium] A dual-socket HPC compute node is equipped with two 32-core Intel Xeon processors running at a base clock frequency of 2.0 GHz. If each core can execute 32 double-precision floating-point operations per

cycle using AVX-512 vector units, what is the theoretical peak double-precision compute performance (Rpeak) of the node?

- A) 1.024 TFLOPS
- B) 2.048 TFLOPS
- C) 8.192 TFLOPS
- D) 4.096 TFLOPS **Answer: D** | *Explanation: Theoretical Peak Rpeak = Sockets (2) * Cores/Socket (32) * Clock (2.0 * 10^9 cycles/sec) * FLOPS/cycle (32) = 4,096 GFLOPS = 4.096 TFLOPS.*

Q03. [Medium] In modern computer architecture and high-performance system design, what fundamental architectural bottleneck is termed the 'Memory Wall'?

- A) The physical DRAM capacity limit imposed by 64-bit operating systems
- B) The growing performance disparity where CPU processing speed grows far faster than main memory (DRAM) bandwidth and access latency
- C) The maximum number of DIMM slots physically permitted on an ATX motherboard
- D) The file size limit imposed by 32-bit legacy file systems **Answer: B** | *Explanation: The Memory Wall refers to the widening divergence between processor compute throughput (which has scaled rapidly via multi-core and vectorization) and DRAM access latency/bandwidth, causing cores to spend significant cycles stalled waiting for data.*

Q04. [Medium] In a production HPC cluster environment, what is the crucial operational distinction between a Login (Head) Node and Compute Nodes?

- A) Login nodes run heavy distributed deep learning training, while Compute nodes only manage user home directories
- B) Login nodes are dedicated to user authentication, code compilation, and job submission, while Compute nodes are reserved exclusively for parallel workload execution managed by a scheduler
- C) Login nodes contain all cluster GPUs, whereas Compute nodes feature only single-core CPUs
- D) Compute nodes allow direct interactive shell access to all users, whereas Login nodes refuse SSH connections **Answer: B** | *Explanation: Login nodes provide access for code editing, compiling, and submitting batch jobs to Slurm, while Compute nodes are isolated worker resources where heavy parallel simulations execute without interactive contention.*

Q05. [Medium] An HPC optimization engineer profiles a simulation algorithm and finds that 20% of the runtime is strictly sequential (s = 0.20), while 80% is parallelizable (p = 0.80). According to Amdahl's Law, what is the theoretical maximum speedup limit achievable by this algorithm even if executed across an infinite number of processor cores?

- A) 4x speedup
- B) 5x speedup
- C) 10x speedup
- D) Infinite speedup **Answer: B** | *Explanation: Amdahl's Law defines Speedup S(N) = 1 / (s + p/N). As N approaches infinity, S_max = 1 / s = 1 / 0.20 = 5x maximum achievable speedup.*

Q06. [Hard] A researcher scales an atmospheric climate model across a 64-core HPC compute node. If 95% of the scaled execution time is spent in parallel computation (p = 0.95) and 5% is sequential (s = 0.05), what is the scaled speedup achieved according to Gustafson's Law (S(N) = s + p * N)?

- A) 60.85x speedup
- B) 20.00x speedup
- C) 5.00x speedup

- D) 128.00x speedup **Answer: A** | *Explanation: Gustafson's Law formula for scaled speedup is S(N) = s + (p * N). For N=64 cores: S(64) = 0.05 + (0.95 * 64) = 0.05 + 60.80 = 60.85x speedup.*

Q07. [Hard] Why was the High Performance Conjugate Gradients (HPCG) benchmark established as a companion to the traditional High Performance LINPACK (HPL) on the TOP500 list?

- A) HPL measures dense linear algebra compute throughput (high compute-to-memory ratio), whereas HPCG stresses memory bandwidth, latency, and sparse, irregular memory access patterns common in modern scientific simulations
- B) HPL evaluates only CPU clock frequency, while HPCG evaluates GPU video memory display outputs
- C) HPCG is designed strictly for consumer laptops, while HPL is designed only for quantum computers
- D) HPL tests single-threaded Python scripts, while HPCG tests compiled C++ assembly **Answer: A** | *Explanation: HPL achieves 80-90% of theoretical peak FLOPS on dense matrix math. In contrast, HPCG exercises sparse matrix-vector multiplication, unstructured grids, and low compute-to-memory traffic, typically achieving under 5% peak FLOPS, accurately reflecting real-world simulation performance.*

🔷 Module 2: Intel HPC Architecture

Q08. [Easy] What dedicated hardware matrix acceleration engine introduced in 4th Gen Intel Xeon Scalable processors (Sapphire Rapids) delivers accelerated deep learning tensor computation directly on CPU cores?

- A) Intel Software Guard Extensions (Intel SGX)
- B) Intel QuickAssist Technology (Intel QAT)
- C) Intel Advanced Matrix Extensions (Intel AMX)
- D) Intel Dynamic Execution Engine (IDEE) **Answer: C** | *Explanation: Intel AMX (Advanced Matrix Extensions) is a built-in silicon accelerator on 4th and 5th Gen Xeon Scalable CPUs that accelerates matrix multiplication operations for INT8 and BF16 deep learning training and inference.*

Q09. [Easy] The Intel Data Center GPU Max Series (formerly codenamed Ponte Vecchio) engineered for exascale supercomputing and deep learning is built upon which GPU microarchitecture?

- A) Intel Xe-LP (Low Power)
- B) Intel Xe-HPG (High Performance Gaming)
- C) Intel Gen9 Graphics
- D) Intel Xe-HPC (High Performance Computing) **Answer: D** | *Explanation: The Intel Data Center GPU Max Series is designed specifically for extreme supercomputing, HPC simulation, and AI training workloads using the Intel Xe-HPC microarchitecture.*

Q10. [Medium] A Habana Gaudi 2 AI accelerator card integrates 24 dedicated 100 Gigabit Ethernet (100GbE) RoCE v2 ports directly on the silicon for scale-out inter-chip communication. What is the total aggregate networking throughput provided per accelerator in Gigabytes per second (GB/s)?

- A) 100 GB/s
- B) 200 GB/s
- C) 600 GB/s
- D) 300 GB/s **Answer: D** | *Explanation: Total throughput = 24 ports * 100 Gbps = 2400 Gigabits per second. Converting to Gigabytes per second: 2400 Gbps / 8 bits/Byte = 300 GB/s aggregate interconnect bandwidth.*

Q11. [Medium] A dual-socket 4th Gen Intel Xeon Scalable server (Sapphire Rapids) features 8 DDR5 memory channels per socket operating at DDR5-4800 (4800 MT/s). Given a standard 64-bit (8-byte) bus width per channel, what is the theoretical peak aggregate memory bandwidth of the dual-socket system?

- A) 153.6 GB/s

- B) 307.2 GB/s
- C) 614.4 GB/s
- D) 1228.8 GB/s **Answer: C** | *Explanation: Theoretical Memory Bandwidth = Sockets (2) * Channels per socket (8) * Transfer Rate (4.8 GT/s) * Bus Width (8 Bytes) = 16 * 4.8 * 8 = 614.4 GB/s aggregate bandwidth.*

Q12. [Medium] Intel Advanced Matrix Extensions (AMX) architecture provides 8 dedicated 2D tile registers (tmm0 through tmm7). If each tile register can configure up to 16 rows of 64 bytes each, what is the maximum total data capacity stored across all 8 AMX tile registers simultaneously?

- A) 512 bytes
- B) 2048 bytes (2 KB)
- C) 4096 bytes (4 KB)
- D) 8192 bytes (8 KB) **Answer: D** | *Explanation: Each AMX tile holds 16 rows * 64 bytes = 1024 bytes (1 KB). Across all 8 tile registers (tmm0 to tmm7), the total capacity is 8 * 1024 bytes = 8192 bytes (8 KB).*

Q13. [Hard] The Intel Xeon CPU Max Series (Sapphire Rapids with HBM2e) integrates up to 64GB of High Bandwidth Memory directly on-package. Which three memory operating modes can be configured via system BIOS?

- A) Striped Mode, Mirror Mode, and Parity Mode
- B) HBM-only Mode, Flat Mode (1LM), and Cache Mode (2LM)
- C) Dynamic Mode, Static Mode, and Swap Mode
- D) Synchronous Mode, Asynchronous Mode, and Dual-Channel Mode **Answer: B** | *Explanation: Xeon Max supports three memory modes: (1) HBM-only (runs with zero external DDR5), (2) Flat mode / 1LM (HBM and DDR5 exposed as distinct NUMA nodes), and (3) Cache mode / 2LM (HBM acts as a transparent hardware-managed L4 cache for standard DDR5 capacity).*

Q14. [Hard] On a 4th Gen Xeon core, AVX-512 VNNI computes up to 256 INT8 operations per cycle, while the Intel AMX TMUL accelerator computes up to 2048 INT8 operations per cycle. By what factor (speedup ratio) does AMX outperform AVX-512 in raw INT8 matrix multiply throughput per clock cycle?

- A) 8x throughput boost
- B) 4x throughput boost
- C) 2x throughput boost
- D) 16x throughput boost **Answer: A** | *Explanation: Throughput ratio = 2048 INT8 ops/cycle (AMX) / 256 INT8 ops/cycle (AVX-512) = 8.0x throughput improvement per clock cycle per core.*

🔷 Module 3: Linux Basics & VFS

Q15. [Easy] In the Linux Virtual File System (VFS) hierarchy, what character designates the single root directory from which all directories, files, and mounted file systems originate?

- A) Backslash ('')
- B) Tilde ('~')
- C) Forward slash ('/')
- D) Hash symbol ('#') **Answer: C** | *Explanation: In Unix/Linux, the single forward slash '/' represents the top-level root directory of the single hierarchical filesystem tree.*

Q16. [Easy] Which standard Linux shell command outputs the absolute canonical pathname of the current active working directory?

- A) 'cd'

- B) 'pwd'
- C) 'whoami'
- D) 'path' **Answer: B** | *Explanation: 'pwd' stands for 'Print Working Directory' and writes the full absolute pathname from root to the current directory.*

Q17. [Medium] In Linux octal permission notation, a systems administrator issues the command 'chmod 750 /scratch/project'. What exact read (r), write (w), and execute (x) permissions are assigned to the Owner, Group, and Others?

- A) Owner: rwx (7), Group: r-x (5), Others: --- (0)
- B) Owner: rw- (6), Group: rw- (6), Others: --x (1)
- C) Owner: r-x (5), Group: rwx (7), Others: r-- (4)
- D) Owner: rwx (7), Group: rwx (7), Others: r-x (5) **Answer: A** | *Explanation: Octal values: 7 = 4(r)+2(w)+1(x) = rwx; 5 = 4(r)+0(w)+1(x) = r-x; 0 = 0(r)+0(w)+0(x) = ---. Therefore 750 grants Owner rwx, Group r-x, and Others no permissions.*

Q18. [Medium] If a Linux user's shell session has an active 'umask' setting of 027, what will be the default octal permissions assigned to a newly created directory (base 777) and a newly created regular text file (base 666) respectively?

- A) Directory: 755 ('rwxr-xr-x'), File: 644 ('rw-r--r--')
- B) Directory: 700 ('rwx------'), File: 600 ('rw-------')
- C) Directory: 770 ('rwxrwx---'), File: 660 ('rw-rw----')
- D) Directory: 750 ('rwxr-x---'), File: 640 ('rw-r-----') **Answer: D** | *Explanation: Base permissions: Directory = 777, File = 666. Applying umask 027 (bitwise bit clear): Directory = 777 - 027 = 750 ('rwxr-x---'). File = 666 masked by 027 = 640 ('rw-r-----'). Group gets r-x (dir) or r-- (file); others get zero rights.*

Q19. [Medium] In a Linux ext4 filesystem configured with the standard 4096-byte (4 KB) allocation block size, if a simulation outputs a configuration file of exactly 10,000 bytes, how many filesystem blocks are allocated on disk and how much internal fragmentation (slack space) is unused in the final block?

- A) 2 blocks allocated, 0 bytes slack space
- B) 3 blocks allocated, 2288 bytes slack space
- C) 4 blocks allocated, 6384 bytes slack space
- D) 1 block allocated, 5904 bytes slack space **Answer: B** | *Explanation: Blocks allocated = ceiling(10,000 / 4096) = 3 blocks (total disk space = 3 * 4096 = 12,288 bytes). Internal fragmentation (slack space) = 12,288 - 10,000 = 2,288 bytes.*

Q20. [Hard] What is the fundamental architectural distinction between the virtual filesystems '/proc' and '/sys' in the Linux kernel?

- A) '/proc' provides a view of process execution state and kernel runtime statistics (procfs), while '/sys' exposes the kernel's unified device model, hardware buses, and driver topology (sysfs)
- B) '/proc' is stored permanently on an NVMe SSD, while '/sys' exists only in CPU cache
- C) '/proc' is read-only for root, while '/sys' can be modified by any unprivileged user
- D) '/proc' manages network routing tables, while '/sys' handles user password hashing **Answer: A** | *Explanation: '/proc' (procfs) exposes active process metadata and runtime kernel state ('/proc/cpuinfo', '/proc/meminfo'), while '/sys' (sysfs) represents the structural hierarchy of physical devices, kernel objects, PCIe busses, and device drivers.*

Q21. [Hard] When a user executes 'rm -f large_dataset.bin' on a Linux filesystem, what happens to the underlying storage data blocks if another running application still holds an open file descriptor to that file?

- A) The data blocks are immediately overwritten with zeros, causing the running application to crash with a segmentation fault

- B) The directory entry is unlinked immediately, but the inode and data blocks remain allocated and accessible to the running application until its file descriptor is closed
- C) The command fails with an 'I/O Resource Busy' error and terminates
- D) The file is automatically relocated to the '/tmp/swap' directory **Answer: B** | *Explanation: In Linux VFS semantics, 'rm' unlinks the directory pointer (reducing the inode's hard-link count). However, disk blocks are only marked free for reuse when the link count reaches zero AND all open file descriptors referencing that inode are closed.*

🔷 Module 4: Shell Scripting

Q22. [Easy] In a Linux Bash automation script, what is the exact function of the initial line '#!/bin/bash' (commonly termed the Shebang)?

- A) It elevates the script's execution privileges to root automatically
- B) It instructs the kernel's 'execve' system call which interpreter binary to launch to parse and execute the script
- C) It compiles the script into an ELF binary executable before running
- D) It marks the entire script file as read-only **Answer: B** | *Explanation: The shebang '#!' is a 2-byte magic number (0x23 0x21) recognized by the Linux kernel loader to invoke the specified interpreter binary ('/bin/bash') with the script file path as an argument.*

Q23. [Easy] Which shell redirection operator appends the Standard Output (stdout) of a command to the end of a specified destination file without overwriting its existing contents?

- A) '>'
- B) '<'
- C) '|'
- D) '>>' **Answer: D** | *Explanation: '>>' opens the destination file in append mode (O_APPEND), adding output to the end of the file. '>' truncates and overwrites existing contents.*

Q24. [Medium] In a Bash automation script, what integer result is printed to the terminal when executing the arithmetic evaluation command: 'echo $(((15 % 4) + (2 ** 5) - (12 / 3)))'?

- A) 27
- B) 42
- C) 35
- D) 31 **Answer: D** | *Explanation: Evaluating step by step according to standard precedence: (15 % 4) = 3; (2 ** 5) = 32; (12 / 3) = 4. Total = 3 + 32 - 4 = 31.*

Q25. [Medium] In production HPC automation scripts, what is the effect of placing the directive 'set -e' at the beginning of the script?

- A) The shell will immediately exit if any command or pipeline returns a non-zero exit status, preventing cascading errors
- B) The shell prints every command to the terminal before executing it
- C) The shell ignores all syntax errors and proceeds execution
- D) The shell allocates maximum GPU memory to all child processes **Answer: A** | *Explanation: 'set -e' (or 'set -o errexit') ensures defensive execution by causing the shell script to abort immediately when a command fails, preventing silent cascading failures in automated batch workflows.*

Q26. [Medium] An HPC batch simulation job terminates unexpectedly. In the shell, checking 'echo $?' yields an exit code of 137. By standard Linux convention, a process killed by an external operating system signal returns '128 + Signal Number'. Which Linux signal terminated the process?

- A) Signal 1 (SIGHUP)
- B) Signal 2 (SIGINT)
- C) Signal 9 (SIGKILL / Out-Of-Memory Killer)
- D) Signal 15 (SIGTERM) **Answer: C** | *Explanation: Signal number = Exit status - 128 = 137 - 128 = 9. Signal 9 is SIGKILL, the uncatchable termination signal typically issued by the Linux kernel OOM (Out Of Memory) killer when RAM limits are exceeded.*

Q27. [Hard] A student writes an automation pipeline: "cat input.dat | grep 'CRITICAL_METRIC' | awk '{print 2}' > output.txt". The input file 'input.dat' does not exist, yet the bash script continues execution and reports exit status '? = 0'. Why did this silent failure occur, and how should it be prevented?

- A) By default, a shell pipeline returns the exit status of the LAST command in the pipeline ('awk'), masking the failure of 'cat'; enabling 'set -o pipefail' ensures the pipeline returns non-zero if ANY stage fails
- B) 'cat' always returns exit code 0 even if files are missing
- C) Redirection with '>' forces the shell to ignore all previous errors
- D) 'awk' elevated the execution privileges to root **Answer: A** | *Explanation: Without 'pipefail', a pipeline's exit status is solely that of the last command ('awk'), which succeeded with empty input (status 0). 'set -o pipefail' forces the pipeline to return the status of the rightmost command that failed with a non-zero code.*

Q28. [Hard] What is an important technical advantage of using the modern Bash compound conditional construct '[[...]]' over the traditional POSIX test operator '[...]'?

- A) '[...]' is executed by the kernel, whereas '[[...]]' runs in user space
- B) '[[...]]' requires compiling the script with GCC before execution
- C) '[...]' only supports floating-point comparisons, while '[[...]]' supports integers
- D) '[[...]]' is a shell keyword that prevents word splitting and pathname expansion on unquoted variable expansions and supports pattern matching with the '=~' regex operator **Answer: D** | *Explanation: '[[...]]' is built into Bash syntax (a keyword, not a built-in executable like '['/'test'). It does not perform word splitting or glob expansion on unquoted variables, preventing syntax errors when variables are empty or contain spaces, and supports advanced regex matching ('=~').*

🔷 Module 5: Secured Shell (SSH)

Q29. [Easy] What is the standard IANA default network TCP port number on which the OpenSSH daemon ('sshd') listens for incoming secure remote shell connections?

- A) Port 80
- B) Port 443
- C) Port 22
- D) Port 8080 **Answer: C** | *Explanation: Standard IANA port assignment for the Secure Shell (SSH) protocol is TCP port 22.*

Q30. [Easy] When setting up passwordless public-key authentication to an HPC cluster login node, where must the user's PUBLIC key be appended on the remote server?

- A) '~/.ssh/authorized_keys'
- B) '~/.ssh/known_hosts'
- C) '/etc/ssh/ssh_config'

- D) '~~/.bashrc' **Answer: A** | *Explanation: The OpenSSH server verifies incoming client key signatures against the public keys stored in the user's '~~/.ssh/authorized_keys' file on the remote machine.*

Q31. [Medium] Modern HPC clusters recommend 256-bit Ed25519 elliptic-curve SSH keys, which offer equivalent cryptographic security strength to classic 3072-bit RSA keys. By what factor (ratio) is the Ed25519 key size smaller than the equivalent RSA key size?

- A) 4x smaller
- B) 8x smaller
- C) 12x smaller
- D) 24x smaller **Answer: C** | *Explanation: Key size reduction factor = 3072 bits (RSA) / 256 bits (Ed25519) = 12.0x smaller key size, delivering faster handshakes and lower network overhead.*

Q32. [Medium] A user generates an Ed25519 key pair and copies the private key to a new workstation. When running 'ssh -i ~/.ssh/id_ed25519 user@hpc.gla.ac.in', the connection immediately terminates with: 'Permissions 0777 for id_ed25519 are too open'. Which exact command resolves this error?

- A) 'chmod 600 ~/.ssh/id_ed25519'
- B) 'chmod 755 ~/.ssh/id_ed25519'
- C) 'chown root ~/.ssh/id_ed25519'
- D) 'rm -f ~/.ssh/known_hosts' **Answer: A** | *Explanation: OpenSSH enforces strict permission checks. If a private key file is readable by group or others (anything more permissive than '600' or '400'), it prints 'WARNING: UNPROTECTED PRIVATE KEY FILE!' and refuses to use the key. 'chmod 600' grants read/write strictly to the owner.*

Q33. [Medium] A research student transfers a 45 Gigabyte (GB) raw simulation dataset from a local workstation to the cluster storage over a dedicated 1 Gigabit per second (1 Gbps) Ethernet link using 'scp'. Assuming 100% link utilization without protocol overhead, what is the theoretical minimum transfer time in seconds?

- A) 45 seconds
- B) 90 seconds
- C) 180 seconds
- D) 360 seconds **Answer: D** | *Explanation: Total data in bits = 45 GB * 8 bits/Byte = 360 Gigabits (Gb). At a network transfer rate of 1 Gbps: Transfer time = 360 Gb / 1 Gbps = 360 seconds (exactly 6 minutes).*

Q34. [Hard] What critical security vulnerability is introduced when an administrator or user enables SSH Agent Forwarding ('ssh -A') when connecting to an untrusted intermediate bastion host?

- A) Any user with root privileges on the intermediate host can access the forwarded SSH agent's Unix domain socket and authenticate to other remote servers as you without knowing your passphrase
- B) The intermediate host steals your raw private key file and saves it in '/etc/shadow'
- C) Agent forwarding permanently decrypts all data on your local laptop hard drive
- D) It exposes the client machine to denial-of-service via ICMP flood **Answer: A** | *Explanation: SSH Agent Forwarding passes a Unix domain socket ('$SSH_AUTH_SOCK') to the remote host. Anyone with root access on that machine can hijack this socket to request signatures from your local key agent, authenticating as you to other machines. 'ProxyJump' ('-J') is the secure alternative.*

Q35. [Hard] In high-frequency automated HPC scripting environments, what is the architectural benefit of configuring SSH Connection Multiplexing using 'ControlMaster' and 'ControlPath'?

- A) It automatically distributes a single SSH session across multiple physical Ethernet switches
- B) It encrypts SSH traffic twice with independent AES-256 ciphers

- C) It maintains a persistent master TCP socket connection, allowing subsequent SSH/SCP sessions to bypass public-key handshakes and authentication overhead
- D) It prevents compute nodes from communicating with each other **Answer: C** | *Explanation: SSH multiplexing ('ControlMaster auto') keeps an initial master connection open. Subsequent 'ssh' or 'scp' invocations communicate through a local control socket ('ControlPath'), reusing the established connection and reducing connection establishment time from ~500ms to ~5ms.*

🔷 Module 6: Intel oneAPI Toolkit

Q36. [Easy] What is the primary industry objective of the Intel oneAPI programming model?

- A) To replace the Linux operating system kernel with an Intel-proprietary microkernel
- B) To provide an open, cross-architecture, standards-based unified programming framework that abstracts software development across diverse CPUs, GPUs, FPGAs, and AI accelerators
- C) To restrict high-performance software development exclusively to Intel Core desktop processors
- D) To serve as a commercial subscription-only cloud database engine **Answer: B** | *Explanation: oneAPI is an open, cross-industry programming model designed to provide a unified development environment across heterogeneous compute architectures (CPUs, GPUs, FPGAs, and specialized AI hardware) without proprietary vendor lock-in.*

Q37. [Easy] Which shell script must be sourced in an HPC Linux environment to set up the necessary PATH, LD_LIBRARY_PATH, and compiler variables for the Intel oneAPI toolkits?

- A) 'source /opt/intel/oneapi/setvars.sh'
- B) 'bash /opt/intel/oneapi/init.sh'
- C) 'exec start_oneapi'
- D) 'python /opt/intel/run_toolkit.py' **Answer: A** | *Explanation: Sourcing '/opt/intel/oneapi/setvars.sh' initializes all environment variables (compiler paths, library search paths, man pages) required to run oneAPI compilers and libraries in the current shell session.*

Q38. [Medium] Intel oneAPI DPC++ (Data Parallel C++) is based upon and actively contributes to which open industry standard maintained by the Khronos Group?

- A) CUDA
- B) OpenMP
- C) SYCL (Single-source C++ heterogeneous programming standard)
- D) Vulkan GLSL **Answer: C** | *Explanation: Intel DPC++ is built upon the open Khronos SYCL standard, enabling single-source modern C++ code to execute across diverse accelerators from multiple hardware vendors.*

Q39. [Medium] In a SYCL kernel execution, a 1-dimensional computational domain of global size 2048 work-items is launched with a work-group size specified as 'sycl::range<1>(256)'. How many independent work-groups will the SYCL runtime schedule across the hardware compute units?

- A) 4 work-groups
- B) 8 work-groups
- C) 16 work-groups
- D) 32 work-groups **Answer: B** | *Explanation: Number of work-groups = Global Range / Local Work-Group Size = 2048 / 256 = 8 work-groups.*

Q40. [Medium] In SYCL Unified Shared Memory (USM), what is the key functional difference between 'malloc_device' and 'malloc_host' memory allocations?

- A) 'malloc_device' allocates memory strictly in accelerator high-bandwidth memory (inaccessible directly by CPU pointers), whereas 'malloc_host' allocates pinned host memory accessible by both CPU and GPU
- B) 'malloc_device' allocates memory on hard disks, while 'malloc_host' allocates memory on tape drives
- C) 'malloc_device' is limited to 1 Megabyte, while 'malloc_host' is unlimited
- D) 'malloc_host' cannot be read by accelerator GPUs **Answer: A** | *Explanation: 'malloc_device' allocates memory resident on the target accelerator device (offering maximum GPU bandwidth, but CPU cannot dereference device pointers). 'malloc_host' allocates pinned host memory accessible by both CPU and device across PCIe.*

Q41. [Hard] A developer allocates a 2D square matrix of dimension 2048 x 2048 containing double-precision floating-point numbers ('sizeof(double) = 8 bytes') using 'sycl::malloc_shared'. How many Megabytes (MB) of memory does this single allocation request?

- A) 8 MB
- B) 16 MB
- C) 32 MB
- D) 64 MB **Answer: C** | *Explanation: Total bytes = 2048 * 2048 elements * 8 bytes/element = 33,554,432 bytes. In Megabytes: 33,554,432 / (1024 * 1024) = 33,554,432 / 1,048,576 = 32.0 MB.*

Q42. [Hard] When creating a SYCL command queue, what operational difference exists between an in-order queue ('sycl::property::queue::in_order') and the default out-of-order queue?

- A) An in-order queue can only run on single-core CPUs and cannot target GPUs
- B) Out-of-order queues automatically discard failed execution tasks without notification
- C) In-order queues disable all asynchronous memory transfers
- D) An in-order queue executes submitted command groups strictly in the sequential order they are submitted, whereas default out-of-order queues can execute commands concurrently unless explicit SYCL events/dependencies are declared **Answer: D** | *Explanation: By default, SYCL queues are out-of-order, allowing kernels to run concurrently or out of sequence unless directed by event dependencies ('depends_on'). Passing 'sycl::property::queue::in_order' enforces FIFO sequential execution, simplifying debugging.*

🔷 Module 7: Intel HPC Toolkit

Q43. [Easy] Which modern LLVM-based C++ compiler provided in the Intel HPC Toolkit is used to compile high-performance SYCL and modern C++ parallel applications?

- A) 'gcc'
- B) 'icpx'
- C) 'javac'
- D) 'cl.exe' **Answer: B** | *Explanation: In the Intel oneAPI HPC Toolkit, 'icpx' is the Intel oneAPI DPC++/C++ compiler, built upon LLVM and providing advanced vectorization and SYCL heterogeneous compilation.*

Q44. [Easy] Which high-performance Fortran compiler in the Intel HPC Toolkit is the modern LLVM-based replacement for the classic 'ifort' compiler?

- A) 'ifx'
- B) 'gfortran'
- C) 'f90'

- D) 'flang' **Answer: A** | *Explanation: 'ifx' is Intel's modern LLVM-based Fortran compiler, delivering modern Fortran standard support and OpenMP offloading to Intel GPUs.*

Q45. [Medium] When compiling computational physics or machine learning C/C++ codes with 'icpx', what optimization does the flag '-xHost' instruct the compiler to perform?

- A) It binds the compiled binary to execute exclusively on a specific network IP address
- B) It runs the code inside an isolated Docker container
- C) It instructs the compiler to generate SIMD vector instructions targeting the highest instruction set supported by the compilation host CPU (e.g., AVX-512)
- D) It disables all loop unrolling and optimization to prioritize compilation speed **Answer: C** | *Explanation: '-xHost' queries the CPU where compilation is occurring and instructs the compiler to generate the most advanced SIMD code available (such as AVX-512 or AVX2), maximizing vector arithmetic throughput.*

Q46. [Medium] An Intel MPI simulation is submitted across 4 cluster compute nodes with the execution command: 'mpirun -n 64 -ppn 16 ./simulation_app'. How many total MPI ranks (processes) are spawned in total, and how many MPI ranks execute per physical node?

- A) 16 total ranks, 4 ranks per node
- B) 32 total ranks, 8 ranks per node
- C) 128 total ranks, 32 ranks per node
- D) 64 total ranks, 16 ranks per node **Answer: D** | *Explanation: Total ranks = 64 (specified by '-n 64'). Processes per node = 16 (specified by '-ppn 16'). Across 4 nodes: 4 * 16 = 64 total MPI processes.*

Q47. [Medium] Which diagnostic analysis tool included in the Intel HPC Toolkit is designed to detect dynamic memory corruption, memory leaks, uninitialized memory accesses, and threading data races without needing specialized recompilation?

- A) Intel Inspector
- B) Intel Advisor
- C) Intel Trace Analyzer and Collector (ITAC)
- D) Intel oneDNN **Answer: A** | *Explanation: Intel Inspector is a dynamic memory and thread checking tool that identifies subtle memory leaks, dangling pointers, deadlocks, and race conditions in multi-threaded C/C++/Fortran programs.*

Q48. [Hard] An HPC engineer profiles an OpenMP simulation on a dual-socket 64-core Intel Xeon node and observes severe memory latency spikes due to cross-socket UPI traffic. Which command pins the process to CPU cores on Socket 0 and restricts physical memory allocation exclusively to Socket 0's local NUMA domain?

- A) Remote NUMA access causes immediate hardware watchdog reboot
- B) numactl --cpunodebind=0 --membind=0 ./simulation_app
- C) nice -n -20 ./simulation_app
- D) nohup ./simulation_app & **Answer: B** | *Explanation: 'numactl --cpunodebind=0 --membind=0' enforces execution strictly on CPU cores belonging to NUMA node 0 and allocates all physical pages from node 0's local memory controller, completely eliminating remote cross-socket UPI bus traversals.*

Q49. [Hard] Hardware profiling on a dual-socket Intel Xeon server indicates that local NUMA memory read access on Socket 0 takes 75 nanoseconds (ns), while accessing remote memory attached to Socket 1 across the Intel UPI interconnect takes 135 nanoseconds (ns). What is the percentage increase in memory access latency incurred by the remote NUMA access?

- A) 20% increase
- B) 45% increase

- C) 80% increase
- D) 150% increase **Answer: C** | *Explanation: Latency increase = (Remote Latency - Local Latency) / Local Latency = (135 ns - 75 ns) / 75 ns = 60 ns / 75 ns = 0.80 = 80.0% latency penalty.*

Q50. [Hard] What is the architectural role of the Open Fabrics Interfaces (OFI / Libfabric) provider layer underneath the Intel MPI Library in high-performance computing clusters?

- A) It acts as a modular, hardware-independent network abstraction framework allowing Intel MPI to interface directly with high-speed kernel-bypass fabrics (InfiniBand verbs, RoCE, Omni-Path, TCP) with minimal overhead
- B) It converts MPI communication messages into HTTP REST API queries
- C) It encrypts cluster network packets using software RSA-4096 keys
- D) It assigns dynamic IP addresses to compute nodes using DHCP **Answer: A** | *Explanation: Libfabric (OFI) is the core fabric interface in Intel MPI. It provides a standard, low-latency API directly above user-space network drivers (like InfiniBand verbs, Cisco usNIC, Intel PSM3, or sockets), enabling kernel-bypass zero-copy communication across HPC fabrics.*